

CBSE Class 12 Physics — Important Derivations

A chapter-wise checklist of the derivations most likely to appear in the CBSE Class 12 Physics board exam — the ones that repeat, year after year, as 2-mark, 3-mark, or 5-mark questions. This is a **list to work through**, not the derivations themselves — the idea is to use it as a revision tracker: can you reproduce each one on paper, from the starting principle, without looking it up?

Note on currency: this list has been checked line by line against the **official CBSE Physics (Code 042) Class XI–XII curriculum for 2026–27** (cbseacademic.nic.in, Chapters 1–14 of the Class XII theory syllabus), not just general search results. Several derivations that used to be standard are no longer in the syllabus at all — Potentiometer, Cyclotron, Velocity Selector, the Toroid, the magnetic dipole moment of a revolving electron, Earth's Magnetism, Eddy Currents, and Brewster's Law — and several more are explicitly downgraded by the syllabus itself to "*no derivation, formula only*" or "*qualitative treatment only*"; those are called out separately below rather than listed as derivations. CBSE still revises this syllabus most years, so re-check against that year's official PDF before finalising a revision plan.

Frequency key: ★★★ = asked almost every year, in some form · ★★ = asked frequently · ★ = asked occasionally, still worth knowing.

Status tags used below: a plain table row is a full derivation you should be able to reproduce.

Formula only means the syllabus explicitly says no derivation is required — know and apply the formula, don't derive it. **Qualitative only** means no maths is expected at all — understand the idea and be able to explain it in words/diagram.

Unit 1 — Electrostatics

Derivation	Typical marks	Frequency
Electric field due to a point charge (from Coulomb's law)	2	★★
Electric field on the axial line of an electric dipole	3	★★★
Electric field on the equatorial line of an electric dipole	3	★★★
Torque on an electric dipole in a uniform external electric field	2–3	★★★
Electric field due to an infinite plane sheet of charge (Gauss's law)	3	★★★
Electric field due to a uniformly charged thin spherical shell — inside, on, and outside the surface (Gauss's law)	3–5	★★★

Derivation	Typical marks	Frequency
Electric field due to an infinitely long straight charged wire (Gauss's law)	3	★★
Electric potential due to a point charge	2	★★
Electric potential due to an electric dipole, at a general point	3	★★
Relation between electric field and potential ($E = -dV/dr$)	2	★★
Electrical potential energy of a system of two point charges	2	★★
Potential energy of an electric dipole in a uniform electric field	2–3	★★
Capacitance of a parallel plate capacitor (no dielectric)	3	★★★★
Capacitance of a parallel plate capacitor with a dielectric slab	3–5	★★★★
Effective capacitance of capacitors in series and in parallel	2–3	★★

Formula only: Energy stored in a capacitor — $U = \frac{1}{2}CV^2 = \frac{1}{2}QV = Q^2/2C$. The syllabus explicitly says "*no derivation, formulae only*" — know the three equivalent forms and when to use each, don't derive it.

Unit 2 — Current Electricity

Derivation	Typical marks	Frequency
Relation between current and drift velocity ($I = nAeV_d$)	3	★★★★
Ohm's law from the microscopic (drift velocity) model; resistivity $\rho = m/(ne^2\tau)$	3–5	★★★★
Balance condition of the Wheatstone bridge	2–3	★★

Not in the current syllabus: the potentiometer (principle, comparing EMFs, internal resistance) has been dropped entirely — it doesn't appear anywhere in the 2026–27 Current Electricity chapter or in the practicals list. Don't spend revision time on it unless your school's internal notes specifically bring it back.

Unit 3 — Moving Charges and Magnetism

Derivation	Typical marks	Frequency
Magnetic field at the centre of a current-carrying circular loop (Biot–Savart law)	3	★★★★
Magnetic field on the axis of a current-carrying circular loop	3–5	★★★★

Derivation	Typical marks	Frequency
Magnetic field due to a straight current-carrying conductor (Biot–Savart law)	3	★★
Force between two long parallel current-carrying conductors; definition of the ampere	3	★★★
Torque on a current-carrying loop in a uniform magnetic field; principle of a moving-coil galvanometer	3–5	★★★
Conversion of a galvanometer into an ammeter (shunt) and a voltmeter (series resistance)	2–3	★★

Qualitative only: the syllabus explicitly restricts several items here to qualitative treatment, no maths expected — know the result and be able to sketch/explain it, but don't prepare a derivation:

- the magnetic field **inside a straight solenoid** (via Ampere's circuital law) — many guides still list this as a full derivation, but the 2026–27 syllabus tags it "*only qualitative treatment*";
- a bar magnet as an **equivalent solenoid**;
- the magnetic field intensity of a dipole (bar magnet) on its **axial and equatorial** lines;
- the **torque on a magnetic dipole** (bar magnet) in a uniform magnetic field.

Unit 4 — Electromagnetic Induction & Alternating Current

Derivation	Typical marks	Frequency
Motional EMF (from the force on a moving charge)	3	★★★
Self-inductance of a long solenoid	3	★★★
Mutual inductance of two long coaxial solenoids	3–5	★★
AC generator — derivation of the instantaneous EMF ($e = e_0 \sin \omega t$) from a coil rotating in a magnetic field	3	★★★
Relation between the RMS and peak values of alternating current/voltage ($I_{\text{rms}} = I_0/\sqrt{2}$)	2	★★
AC through a pure resistor / pure inductor / pure capacitor — current, phase relationships	2–3 each	★★
Impedance of a series LCR circuit using a phasor diagram	5	★★★
Resonance condition and resonant frequency of a series LCR circuit	3	★★
Average power in an AC circuit; power factor	2–3	★★

Derivation	Typical marks	Frequency
Working and EMF equation of a transformer ; efficiency	3	★★

Electromagnetic Waves (Chapter 8) isn't given its own numbered section below — the entire chapter (displacement current, wave characteristics, transverse nature, the EM spectrum) is syllabus-tagged "*qualitative idea only*", so there's nothing to derive from it.

Unit 5 — Ray Optics & Optical Instruments

Derivation	Typical marks	Frequency
Mirror formula for a concave mirror (real image case)	3	★★
Relation between critical angle and refractive index for total internal reflection ($n = 1/\sin i_c$)	2	★★
Refraction at a single spherical surface — relation between u , v , R	3–5	★★★★
Lens maker's formula	3–5	★★★★
Thin lens formula ($1/v - 1/u = 1/f$)	2–3	★★
Equivalent focal length/power of two thin lenses in contact	2–3	★★
Refraction through a prism — $i + e = A + \delta$, and $n = \sin[(A+D_m)/2] / \sin(A/2)$	3–5	★★★★
Magnifying power of a simple microscope	2–3	★★
Magnifying power of a compound microscope (normal adjustment)	3–5	★★★★
Magnifying power of an astronomical telescope (normal adjustment)	3–5	★★★★

Unit 6 — Wave Optics

Derivation	Typical marks	Frequency
Proof of the laws of reflection and refraction using Huygens' principle	3	★★★★
Conditions for constructive and destructive interference (path difference)	2–3	★★★★

Formula only: Young's double-slit fringe width, $\beta = \lambda D/d$. The syllabus explicitly says "*expression for fringe width (no derivation, final expression only)*" — know what each symbol means and be able to apply it, don't derive it.

Qualitative only: single-slit diffraction (the pattern, and why the central maximum is twice as wide as the secondary ones) — a mathematical derivation of the fringe/angular width is not expected, only the qualitative explanation.

Unit 7 — Dual Nature of Radiation and Matter

Derivation	Typical marks	Frequency
Einstein's photoelectric equation	3	★★★
de Broglie wavelength of a moving particle	2–3	★★★

Unit 8 — Atoms

Derivation	Typical marks	Frequency
Bohr's postulates → radius of the nth orbit of the hydrogen atom	3–5	★★★
Bohr's model → velocity of the electron in the nth orbit	2–3	★★
Bohr's model → total energy of the electron in the nth orbit	3–5	★★★

Qualitative only: the hydrogen line spectra (Lyman, Balmer, Paschen series, etc.) and the Rydberg formula are listed as qualitative treatment only — know the series and what the formula predicts, but a derivation of the formula itself isn't expected.

Unit 9 — Nuclei

No standalone derivations in the current syllabus. Chapter 13 (Nuclei) now covers only the composition and size of the nucleus, nuclear force, the mass-energy relation and mass defect, binding energy per nucleon and its variation with mass number, and nuclear fission and fusion — all conceptual, not derivation-based. The radioactivity topic (decay law $N = N_0 e^{(-\lambda t)}$, half-life, and mean life) has been **removed from the syllabus entirely** and should not be prepared as a derivation, or at all, unless your school's internal notes say otherwise.

Unit 10 — Semiconductor Electronics

Mostly conceptual in the current syllabus (V–I characteristics, working principles, logic gates) — no major standalone derivations are typically asked from this unit, unlike Units 1–9 above.

How to use this list

1. Go unit by unit. For each derivation, close the book and try to produce it from the starting principle (Coulomb's law, Gauss's law, Biot–Savart law, Snell's law, Bohr's postulates, etc.) through to the final result, with the diagram.
2. Mark anything you couldn't do cleanly — those go on a short list for a second pass closer to the exam.
3. The ★★★ items are worth over-preparing: they show up almost every year, sometimes as the derivation itself, sometimes as a numerical built on top of it.
4. Board papers often ask you to **derive, then apply** (e.g. derive the lens maker's formula, then use it in a numerical) — so pair each derivation with a couple of numericals on the same formula while revising.

Sources: the official **CBSE Physics (Code 042) Class XI–XII curriculum, 2026–27**, published by cbseacademic.nic.in — every Chapter 1–14 content line for Class XII theory was checked directly against it, which is why items like the potentiometer, the solenoid field, and the radioactivity topic are now handled exactly the way that document states, rather than on general knowledge of past years' patterns. Frequency ratings (★) still come from recent-year CBSE sample papers and previous-year question papers, since the syllabus document itself doesn't rank importance.